CN and CH Molecular Band Strengths in the Peculiar Open Cluster NGC 6791

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Motivation

- NGC 6791 is not a typical open cluster (OC)

	Typical OC	NGC 6791[1]	Typical GC
Mass ($M_\odot$)	10^2-10^3	5 x 10^4	10^5-10^6
[Fe/H]	metal-rich	+ 0.4	metal-poor
Age	wide range	8 Gyr	12-13.5 Gyr

- Photometric[2] and spectroscopic[3] studies have found evidence of an extended period of star formation and a Na-O anti-correlation in the cluster
- Molecular band strengths have also shown evidence of multiple populations[4]
- These are not features one would expect to find in a Galactic OC
- The origin and evolution of NGC 6791 are still being debated and not well constrained
- We are using data from the Sloan Extension for Galactic Understanding and Exploration (SEGUE) to measure the CN and CH molecular band strengths of cluster members to test for the presence of chemical inhomogeneities in the cluster

Data

- Spectra were collected to validate the SEGUE Stellar Parameter Pipeline
- Low resolution R ~ 2000
- Wavelength range: 3600 – 9200 Å
- Strict membership criteria
- 76 Cluster members
- 57 Main Sequence stars
- 13 Lower Red Giant Branch (LRGB) stars
- 6 Red Clump (RC) stars

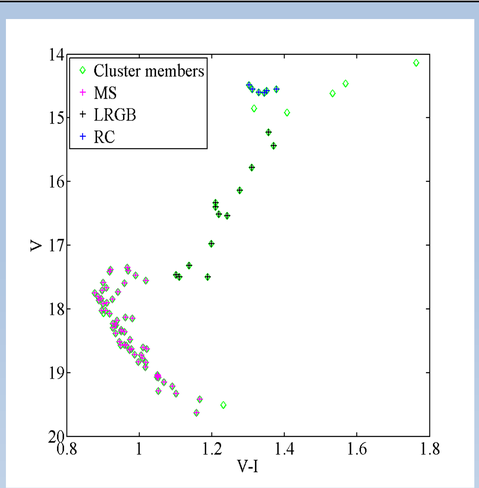

Figure 1. V vs. V-I color magnitude diagram for NGC 6791. The different evolutionary stages in the cluster have been plotted with different colored crosses.

CN and CH Molecular Band Strengths

- Each band strength is defined as a magnitude[5]

$$-2.5\log\frac{\int \text{Flux(Spectral region } \textbf{with} \text{ CN/CH feature)}}{\int \text{Flux(Nearby region } \textbf{without} \text{ feature)}}$$

- These spectral regions are illustrated on a cluster LRGB spectrum in the following figure

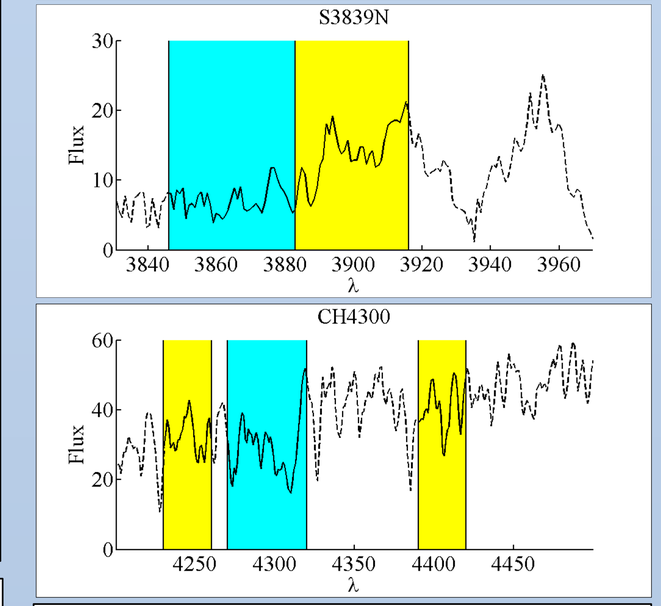

Figure 2. The spectral windows for measuring the molecular band strengths. The blue regions contain the CN or CH feature, and the yellow regions do not.

Band Strength Variation

- Band strength depends on:
  - Surface gravity
  - Temperature
  - Chemical composition
- Variation in band strength within luminosity and temperature range of evolutionary stage is expected
- Large variations in CN band strength:
  → variations in composition
- In GC's, CN band strength distribution is bimodal[6]
- OC's typically have a unimodal distribution

Band Strength Error (Monte Carlo Method)

- A noise value is added to each pixel in a given spectrum
- The noise value is drawn from a normal distribution with a FWHM = variance in that pixel
- The noised added spectrum is used to calculate band strengths
- Process repeated 100 times
- Error = **σ** (100 band strength measurements)

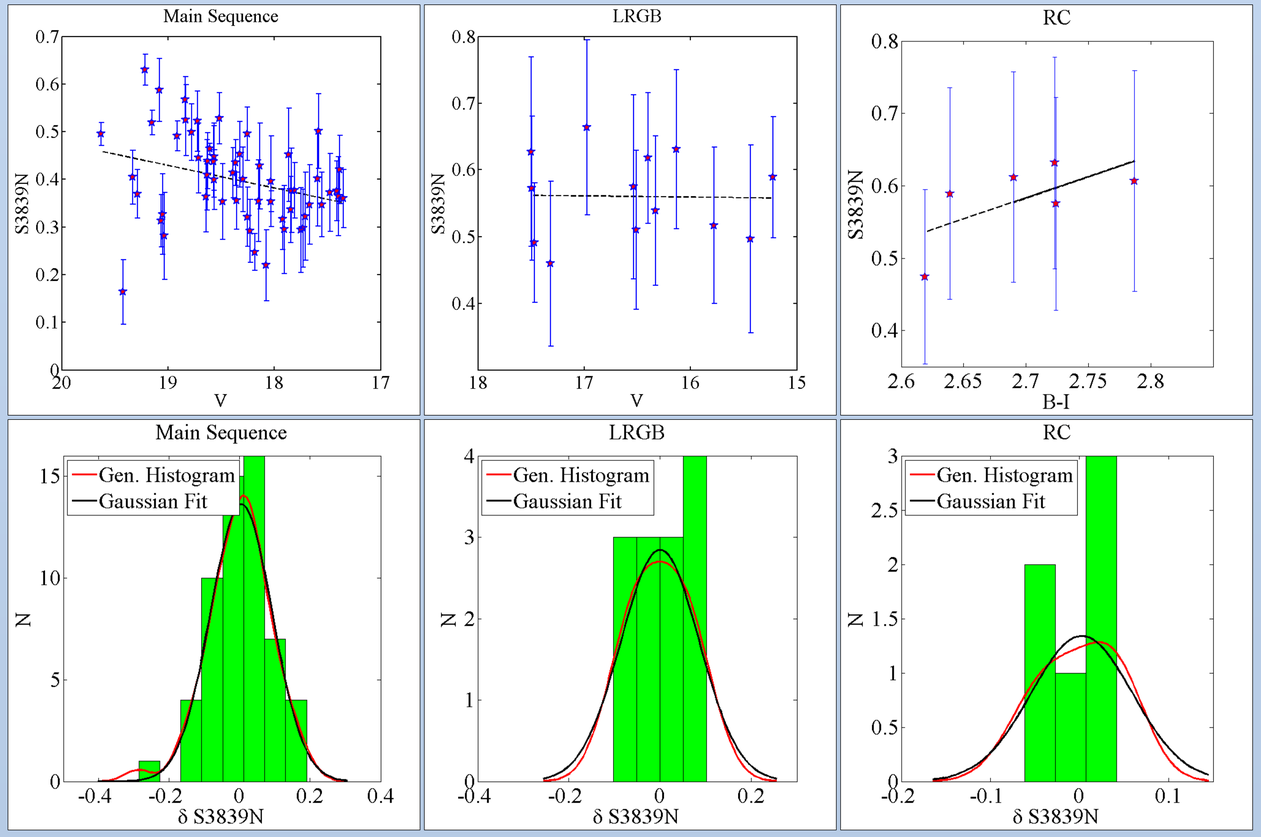

Figure 3. The $S3839_N$ band strengths plotted versus V magnitude, or color (for clump stars), are shown in the 1st row of the figure. A line is fit to the data in each plot to determine the dependence of band strength on luminosity or color within each evolutionary stage. A pseudo index, denoted by a δ, is determined by subtracting the band strength of each star from the linear of fit. This removes the effects of luminosity variations within a given evolutionary stage. The second row plots the distributions of the pseudo index for $S3839_N$. This analysis was also performed for the CH4300 and S4141 indices.

CH4300	error	σ	S3839N	error	σ	S4142	error	σ
MS	0.0288	0.0170	MS	0.0679	0.0880	MS	0.0186	0.0205
LRGB	0.0319	0.0154	LRGB	0.1191	0.0858	LRGB	0.0541	0.0218
RC	0.0197	0.0095	RC	0.1462	0.0569	RC	0.0477	0.0126

Results & Discussion

- **error**: median error on index measurement for stars in each evolutionary stage
- **σ**: standard deviation of Gaussian fit to generalized histogram
- If $\boldsymbol{\sigma > error}$:
  → Band strength distribution is wider than expected from the error in the measurement
  → Variation in **chemical composition** of cluster members
- In all but one case (S3839N MS stars) $\boldsymbol{\sigma < error}$
- No dispersions in CN or CH band strength beyond what is expected from observational error
- There were no correlations of δ between indices → residuals are not correlated as would be expected with abundance variations
- From these results we can not conclude that there are abundance variations amongst cluster members of NGC 6791.

References:

1. Mass: Platias et al. 2011, Age: Carraro et al. 2006, Metallicity: Carreta et al. 2007
2. Twarog et al. 2011
3. Geisler et al. 2012
4. Carrera 2012
5. S3839: Briley et al. 1990, CH4300: Harbeck et al 2003, S4142: Norris & Freeman 1979
6. Martell et al. 2009

Acknowledgements and contact information:
This work is supported through the NSF Graduate Student Fellowship program.
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